Precision Requirements for GPU-Accelerated Constraint-Based Causal Discovery

Research proposal and experimental plan — working draft

Thesis. Reduced floating-point precision perturbs conditional independence test statistics by an amount governed by the condition number of the inverted correlation submatrix. Because the PC algorithm's control flow is path-dependent, a small per-test decision-flip rate is amplified into a much larger divergence in the recovered graph. This amplification has never been measured, and it is most severe precisely in the time-series setting, where lagged variables of an autocorrelated series are near-collinear by construction.

1. Motivation

Constraint-based causal discovery is increasingly run on GPUs. cuPC (IEEE TPDS, 2020) reports average speedups of roughly 500× and 1300× over serial CPU implementations; Schmidt et al. (SSDBM 2018) report factors up to 729 on gene expression data. Every one of these implementations makes an implicit numerical choice — which floating-point format to use for the partial-correlation computation — and none of them examines it.

The choice is not cosmetic. FP64 throughput differs by roughly 30× between datacenter and consumer GPUs, so "use double precision" is nearly free on an A100 and crippling on an L40S. Meanwhile the algorithm's output is a discrete object: a graph. A rounding difference matters only when it flips a p-value across the significance threshold — but when it does, the consequence is not a small numerical error, it is a different causal conclusion.

1.1 The gap

A systematic search of the GPU causal-discovery literature, the mixed-precision numerical linear algebra literature, and statistical computing found no work measuring how precision affects edge decisions or recovered graph structure in constraint-based causal discovery.

1.2 A false positive in the literature

One secondary source appears to fill this gap. Pothireddi (*World Journal of Advanced Research and Reviews*, 2025, 26(01), 852–866) attributes to Schmidt, Huegle & Uflacker (SSDBM 2018) a detailed precision study with condition-number thresholds, FP32/FP64 graph comparisons, and a hybrid-precision recommendation. Direct inspection of the SSDBM paper shows it contains none of this. The discrepancies are systematic:

Claim attributed to Schmidt et al.	What the SSDBM paper actually contains
Precision experiments; FP32 identical to FP64 below κ = 100; ~2.3% of pairs differ above κ = 1000	No precision comparison and no condition number anywhere in the paper
Benchmarks on an NVIDIA Tesla P100	NVIDIA Titan X, 12 GB, 336 GB/s, 6.14 TFLOPS
650-variable genomic and 1,750-variable financial datasets	Five gene expression datasets (NCI-60, MCC, BR51, S. aureus, S. cerevisiae), 1,190–5,361 variables
Speedups of 60–80× vs. single-threaded CPU	Up to 122× (level 0), 746× (level 1), 729× combined, vs. pcalg
Triangular adjacency storage reducing memory ~50%; sepset structure saving 78%	Triangular storage is named as *future* work; not implemented
Multi-GPU scaling at 86% efficiency across four GPUs	Multi-GPU support listed as an open question for future work
Hybrid precision strategy yielding ~1.6× speedup	Does not appear

The only numerically adjacent statement in the entire SSDBM paper is that when the determinant of the correlation matrix is zero, the Moore–Penrose pseudo-inverse is used — a singularity guard, not an analysis. The paper also restricts itself to levels 0 and 1 of the skeleton search, with higher levels explicitly deferred to

future work.

Positioning language for the paper. "Pothireddi (2025) attributes an FP32/FP64 precision analysis with condition-number thresholds to Schmidt et al. (2018); the cited work contains no such analysis." This is factual and verifiable, requires no claim about intent, and inoculates the contribution against a reviewer who encounters the secondary source.

2. Contributions

- **C1 — Amplification.** Quantify the ratio between the per-CI-test decision-flip rate and the final structural Hamming distance between graphs recovered at different precisions. This ratio is a property of the algorithm's path dependence, not of the arithmetic, and is the paper's central methodological unit.
- **C2 — Condition-number mediation.** Establish κ of the inverted submatrix as the explanatory variable linking format to flip probability, and test the classical digits-lost $\approx \log_{10}\kappa$ heuristic against measured behaviour.
- **C3 — The time-series hazard.** Show that lagged variables of a strongly autocorrelated series are near-collinear by construction, so ill-conditioning arises independently of the sample-to-dimension ratio that governs the classical analysis.
- **C4 — Cross-device reproducibility at matched precision.** Measure whether GPU and CPU implementations agree at identical precision, given differing implementations of the transcendental functions used in the Fisher-z tail.
- **C5 — A practitioner decision procedure.** A cheap pre-flight diagnostic and a mixed-precision policy with measured accuracy-per-throughput trade-offs.

2.1 Two questions, deliberately separated

Q1 (numerical/statistical). Does reduced precision change edge decisions? IEEE-754 FP32 is FP32 regardless of the device, and formats without CPU hardware support (BF16, FP8) are simulated by rounding to the target format and computing in a wider one — the standard approach in numerical analysis. **Q1 therefore requires no GPU at all**, which decouples the scientific contribution from hardware availability and queue time.

Q2 (performance). What does each format cost on which hardware? Entirely device-dependent, and the only part that needs the cluster.

3. What the SSDBM paper contributes to the design

Having obtained the primary source, three findings in it directly shape the experiments.

3.1 The $\alpha = 1.0$ device solves the pairing problem

Schmidt et al. set the significance level to 1.0 to force a worst case in which no edge is removed, giving a fixed test count for scalability measurement. The same device solves a problem in our design: if precision flips a level-0 decision, level 1 operates on a *different test set* and the paired comparison becomes ragged. Running at $\alpha = 1.0$, recording every p-value, and applying thresholds post hoc yields an exactly paired per-test measurement at any α, cleanly separating the raw flip rate from the cascade.

3.2 Level 1 dominates — which changes which matrices matter

Their runtime breakdown shows level 1 consuming 94.6–99.6% of skeleton-discovery time; level 0 accounts for 0.01–0.27% despite being 2–21% of all tests; levels 2 and above never exceed 5.4%. For $|S| = 1$ the inverted matrix is 3×3, not the 8×8 that a naive cost model assumes. Whether autocorrelated telemetry shifts work into higher levels is now an empirical question that must be answered early, because it is the difference between ill-conditioning being marginal and being dominant.

3.3 Two amplification channels beyond the skeleton

Their implementation evaluates all separation-set candidates in parallel rather than stopping at the first acceptance, and preserves candidate ordering by position in the correlation matrix to maintain order-independence. Which candidate is *recorded* as the separating set can therefore shift with precision — and separating sets determine v-structures, hence orientation. Precision can change the CPDAG even when the skeleton is identical. Separately, the determinant-zero pseudo-inverse fallback is a branch condition, so reduced precision can flip which code path executes, not merely perturb a value.

4. Experimental plan

Six tiers. Tier 0 must pass before any result in later tiers is interpretable. Tiers 0–2 plus E4.1 constitute the paper; Tier 3 bridges to the time-series project; Tiers 4–5 are appendix material.

Tier 0 — Controls

ID	Experiment	Purpose / pass criterion
E0.1	Noise floor: FP64 vs. FP64, identical seed, repeated runs, with OMP_NUM_THREADS=1 and OPENBLAS_NUM_THREADS=1.	Flip rate must be exactly zero. Multithreaded BLAS varies reduction order between runs. If this is non-zero, stop — everything downstream measures non-determinism, not precision.
E0.2	Cross-device discrepancy at matched precision: FP64 CPU vs. FP64 GPU. The GPU path uses CUDA's log1p() and normcdf(); the CPU reference uses R's pnorm. At $\alpha = 0.01$ the comparison lives in the distribution tail, where implementations diverge most.	Possibly the strongest single result: GPU causal discovery may be non-reproducible against CPU references *even at identical precision*. Note that Schmidt et al.'s own validation criterion was agreement with a CPU implementation.
E0.3	Algorithmic-variant control at FP64: Fisher-z via log1p vs. naive log((1+r)/(1-r)); partial correlation via explicit inverse vs. Cholesky vs. regression residuals.	Establishes how much variation is attributable to algorithm choice before any of it is attributed to dtype. Likely a larger effect than precision itself.

Tier 1 — Per-test flip rate

ID	Experiment	Measurement
E1.1	Paired p-value sweep at $\alpha = 1.0$ across FP64, FP32, TF32, BF16 and FP8-E4M3 (low formats simulated by rounding, e.g. via ml_dtypes). Fixed test set; thresholds applied post hoc.	Per-test decision-flip rate at $\alpha \in \{0.001, 0.01, 0.05\}$. Exactly paired by construction.
E1.2	Condition-number mediation: log κ for every inverted submatrix using the Cholesky diagonal ratio as a cheap proxy, calibrated against exact SVD on a random subsample.	Flip probability regressed on κ, with the digits-lost $\approx \log_{10}\kappa$ prediction overlaid. Avoid np.linalg.cond in the inner loop — the SVD would dominate runtime.
E1.3	Level stratification: flip rate by conditioning-set size $\lvert S\rvert$.	Report both the raw rate and the rate weighted by each level's runtime share. A high flip rate at level 4 is immaterial if level 4 is 0.5% of the work.

Tier 2 — Amplification (the core result)

ID	Experiment	Measurement
E2.1	Cascade: full skeleton discovery at realistic α, all precisions, paired seeds.	Amplification factor = SHD between precision variants divided by the per-test flip rate from E1.1.
E2.2	Order-dependence interaction: PC-stable vs. order-dependent PC.	Does order-independence dampen amplification, and by how much? A second result in its own right.
E2.3	Separating-set divergence.	Fraction of sepsets differing between precisions, and the resulting CPDAG differences among cases with identical skeletons. This channel is currently unmeasured by anyone.
E2.4	Pseudo-inverse branch rate, sweeping the determinant / rcond threshold.	Frequency of the singularity fallback by precision, reported as a code-path flip distinct from numeric perturbation.

Tier 3 — Time-series specific

ID	Experiment	Measurement
E3.1	Autocorrelation sweep on VAR data: $\phi \in \{0, 0.3, 0.6, 0.9, 0.95, 0.99\}$, $\tau_{max} \in \{1, 3, 5, 10\}$.	κ distribution as $\phi \to 1$, and flip rate as a function of ϕ. This is the mechanism claim (C3): near-collinearity by construction, independent of the sample-to-dimension ratio.
E3.2	Level-distribution shift: replicate the SSDBM test-count and runtime breakdowns for autocorrelated time series.	If autocorrelation pushes work into levels ≥ 2, that alone is publishable — it means the levels-0-and-1 acceleration strategy does not transfer to the time-series setting.

Tier 4 — Harm, and the practitioner deliverable

ID	Experiment	Measurement
E4.1	Harmful vs. harmless divergence, against synthetic ground truth.	SHD-vs-truth per precision, not merely precision-vs-precision. Near-threshold coin flips that do not degrade accuracy are a far softer finding than systematic loss of true edges. Report honestly if a lower precision scores better on some seeds.
E4.2	Directional asymmetry of errors.	Are differences dominated by false negatives (missed edges)? The fabricated secondary source asserts this; it is a plausible guess and worth settling either way.
E4.3	Near-threshold mass as a pre-flight diagnostic.	Fraction of tests with $p \in [\alpha/2, 2\alpha]$ as a predictor of flip risk — a one-line check a practitioner can run before choosing a format.
E4.4	Mixed-precision policy: FP32 screening with FP64 recomputation near threshold.	Recompute fraction required for zero flips, and the resulting net speedup. Connects to the iterative-refinement literature (Carson & Higham; GMRES-IR).

Tier 5 — Throughput (HPC; secondary)

ID	Experiment	Measurement
E5.1	Throughput by precision × GPU class × $\|S\|$ × batch size, across datacenter and consumer parts.	Following SSDBM reporting precedent, include achieved_occupancy and sm_efficiency, and report timings separately with and without host–device transfer.
E5.2	Accuracy-per-throughput Pareto frontier.	The practitioner-facing figure: what each format costs in graph accuracy for what it returns in speed, on each class of hardware.

4.1 Replication arm

The five gene expression datasets used by Schmidt et al. are public and named. Running Tier 1 on exactly their setup allows the paper to state that it conducts the precision analysis which has been erroneously attributed to that work — a memorable framing that grounds synthetic results in an established benchmark and demonstrates the gap rather than asserting it.

Note the regime difference. Their datasets have 47–160 observations against 1,190–5,361 variables; telemetry has $T \approx 10{,}000$. The Fisher-z scaling factor $\sqrt{(n - |S| - 3)}$ differs by an order of magnitude, so near-threshold mass will differ substantially between the two regimes. Reporting both is itself a finding.

5. Implementation

Because Q1 is a question about arithmetic rather than hardware, the scientific contribution can be completed on a laptop. Simulate the target format by rounding to it and computing in a wider one; this yields FP8 results without an H100.

Component	Tool	Hardware
Precision study (Tiers 0–4)	NumPy + ml_dtypes for formats; Numba @njit for the skeleton loop	Laptop
CI-test throughput benchmark	CuPy or PyTorch batched linear algebra	HPC
Custom kernel, only if profiling demands it	Numba CUDA, then CuPy RawKernel	HPC

5.1 Pitfalls that will otherwise invalidate results

- **Single-threaded BLAS.** Set OMP_NUM_THREADS and OPENBLAS_NUM_THREADS to 1 before importing NumPy, or reduction-order variation will contaminate paired comparisons (see E0.1).
- **TF32 masquerading as FP32.** On Ampere and later, PyTorch may silently use TF32 (a 10-bit mantissa) for float32 matmul; the defaults have changed across versions. Pin torch.backends.cuda.matmul.allow_tf32 and torch.backends.cudnn.allow_tf32 explicitly, verify empirically, log the state with every result, and treat TF32 as its own level rather than a contaminant.
- **Silent dtype promotion.** NumPy promotion rules changed in 2.0 (NEP 50). Pin the version and assert dtypes on entry and exit of every stage.
- **Numba parallel=True reintroduces non-determinism** through reduction ordering. Use it for throughput measurement only, never for paired precision runs.
- **Determinism settings.** Fix seeds, library versions, and driver versions; log them alongside every reported number.

6. Risks and kill conditions

Gate	Test	Kill condition
G0	E0.1 noise floor	Non-zero FP64-vs-FP64 flip rate that cannot be eliminated — the measurement apparatus is invalid
G1	E1.1 at $\phi = 0.95$, moderate N	Zero flips at FP32 *and* TF32 — no effect to report at practical precisions; retreat to BF16/FP8 only, or reframe as a negative result
G2	E2.1 amplification	Amplification factor ≈ 1 — the path-dependence claim (C1) collapses and the paper becomes a routine precision table
G3	E3.2 level distribution	Time-series work stays in levels 0–1 with well-conditioned 3×3 matrices — the time-series motivation (C3) weakens considerably

> **A negative result is publishable here.** "Constraint-based causal discovery is robust to FP32 and TF32 across the tested regimes, and the mechanism is that level-1 tests invert well-conditioned 3×3 matrices" is a useful, citable finding. It licenses every GPU implementer to drop to FP32 and gain 30× on consumer hardware. Pre-register this outcome rather than discovering it and quietly abandoning the project.

7. Scope and venue

This is a workshop or short paper and should be time-boxed as one. It will attempt to grow into a full paper; freeze the factor list before starting and write the results-table skeleton first. Candidate venues include causality workshops at NeurIPS or ICML, the CLeaR workshop track, or a reproducibility-oriented venue — the latter

being a good fit if E0.2 produces a real effect.

If cross-device disagreement at matched precision does materialise, consider leading with it. "GPU and CPU causal discovery disagree at identical precision" is a sharper opening than a precision table, and it reframes the contribution around reproducibility rather than speed.

Primary sources

Schmidt, C., Huegle, J., Uflacker, M. (2018). Order-independent constraint-based causal structure learning for Gaussian distribution models using GPUs. *SSDBM '18*, Article 19, 1–10. doi:10.1145/3221269.3221292

Zarebavani, B., Jafarinejad, F., Hashemi, M., Salehkaleybar, S. (2020). cuPC: CUDA-based parallel PC algorithm for causal structure learning on GPU. *IEEE TPDS* 31(3), 530–542.

Colombo, D., Maathuis, M.H. (2014). Order-independent constraint-based causal structure learning. *JMLR* 15, 3741–3782.

Carson, E., Higham, N.J. (2018). Accelerating the solution of linear systems by iterative refinement in three precisions. *SIAM J. Sci. Comput.* 40(2), A817–A847.

Pothireddi, S.C. (2025). GPU optimization for causal AI: accelerating the PC algorithm. *WJARR* 26(01), 852–866. *Cited only as an unreliable secondary source; see §1.2.*